# DevOps

Design
Coding
Test

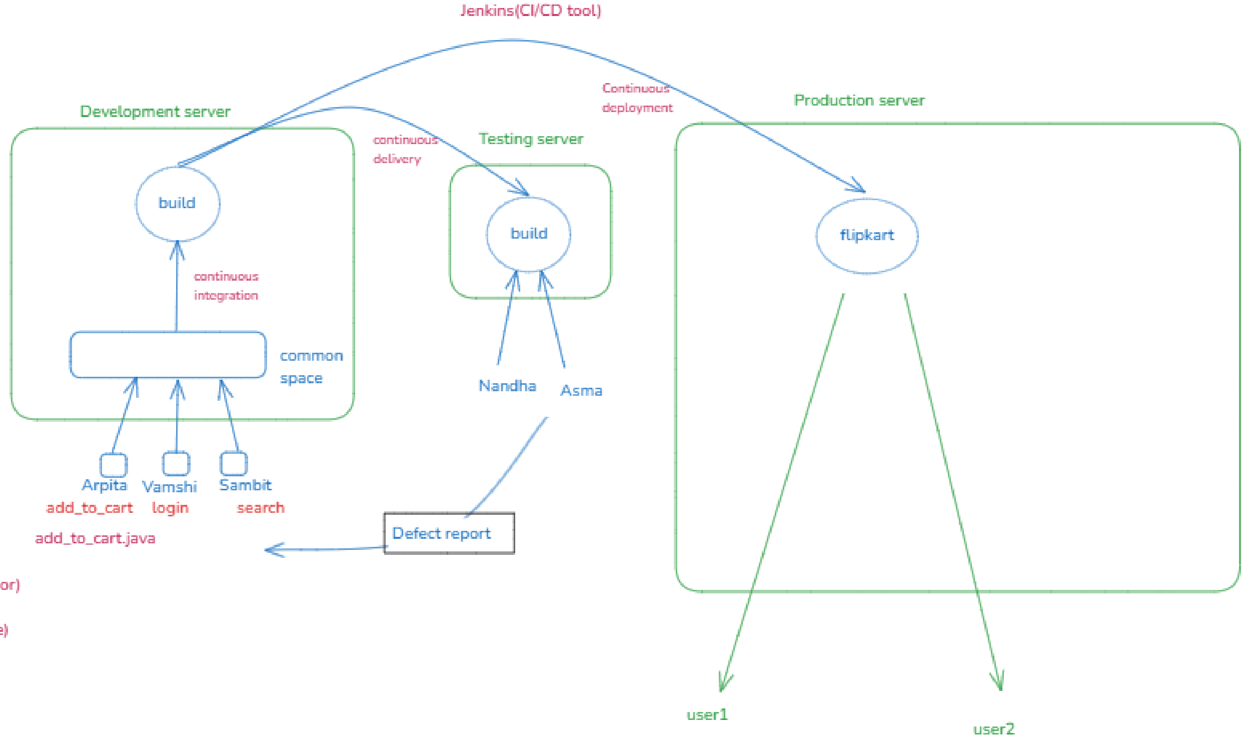
Installing s/w
Maintenance

SDLC

Requirement collection
Feasibility study
Design
Coding
Testing
Installing/deploying in customer's place
Maintaining of server

unblock team number - 8951939012

qtalk team number - 7338471266




Build tool

maven
PyBuilder
NPM
Ant
Gradle
make/cmake

Book

shoe

flipkart

Add to cart
login
search

GitHub

GIT
(VCS tool)

3 w's

Who (author)

when (time)

what

Docker

Kubernetes

Terraform (IAC tool)

Ansible

Linux
Shell scripting

**DevOps:**

It is a process or approach or culture which will enhance the process of development and deployment of an application in the server.

**Benefits:**

1. Productivity will increase.
2. Tools are speeding up the process.
3. Collaboration is increased.
4. Quality of product will be good.
5. Load is decreased.

**Resources**

**Types of Cloud Computing**

Cloud deployment model

1. Public cloud
2. Private cloud
3. Hybrid cloud
4. Community cloud

Cloud service model

1. IAAS(Infrastructure as a service)
2. PAAS(Platform as a service)
3. SAAS(Software as a service)

Modes of transport:

| Public transport(Bus) | Own vehicle(Car/Bike) | Rental Vehicle(Taxi) |
|---|---|---|
| - No entry restriction | - Entry restrictions | - Entry restrictions |
| - Customization is not possible | - Customization is possible | - Customization is not possible |
| - Less privacy | - Privacy | - Privacy |
| - Maintenance is done by company | - Maintenance is done by user/owner | - Maintenance is done by taxi owner |

# Cloud deployment model

**Public cloud:**

- for all general purpose
- default conditions which we can't customize
- Less private
- Maintenance is done by 3rd party cloud providers
- Pay-as-you-go model(PayG)
- deployed globally

**Private cloud:**

- Open only for single organization or a person
- Customization is possible
- Privacy will be there
- Maintenance is done by 3rd party cloud service probiders or company
- Pay-as-you-go model(PayG)
- Deployed locally

**Hybrid cloud:**

- It is a combination of both private and public cloud

**Community cloud:**

- Sharing the cloud with multiple organizations

***Cloud Service Models***

| ON-PREMISES PRIVATE CLOUD | INFRASTRUCTURE AS A SERVICE | PLATFORM AS A SERVICE | SOFTWARE AS A SERVICE |
|---|---|---|---|
| Data & access | Data & access | Data & access | Data & access |
| Applications | Applications | Applications | Applications |
| Runtime | Runtime | Runtime | Runtime |
| Operating system | Operating system | Operating system | Operating system |
| Virtual machine | Virtual machine | Virtual machine | Virtual machine |
| Compute | Compute | Compute | Compute |
| Networking | Networking | Networking | Networking |
| Storage | Storage | Storage | Storage |

**AWS**
**(Amazon web service)**

- AWS is a cloud service provider
- AWS is a subsidiary of Amazon
- AWS was launched in the year 2002
- SQS(Simple queue service) was launched in the year 2004
- In the year 2006, AWS officially launched 2 service for public use
    1. EC2(Elastic compute cloud)
    2. S3(Simple storage service)
- 200+ services are available

## AWS Global Infrastructure:

**Important components:**

1. **Regions**
2. **Availability Zones**
3. **Edge locations**
4. **Regional edge caches**

# AWS

## *Amazon Web Service*

***Amazon Web Services (AWS) was launched in 2006 by Amazon to provide cloud computing services.***


- It offers a wide range of infrastructure services such as computing power, storage, and databases.
- AWS has since become a leader in the cloud industry, powering businesses globally with scalable and flexible solutions.

**Regions:**

- Geographical location
- Cluster of Availability zones(AZ)
- Minimum of 3 AZ, and max. 6 AZ
- 36 regions

**Availability zones:**

- It is a location where actual data centers present within a region
- It is responsible for better network connectivity
- There are currently 114 AZ's

**Edge Locations**

- These are actual data centers present outside the region.
- It is responsible for better network connectivity
- It is used to delivery content with low latency
- These are also called POP's (Point of presence)
- Currently 700+ edge locations

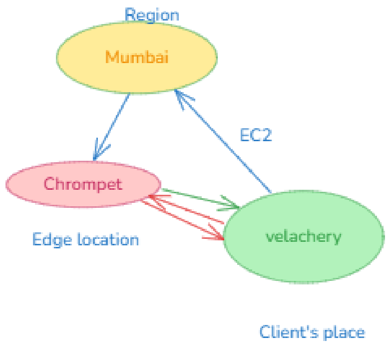
**Regional Edge Caches(REC):**

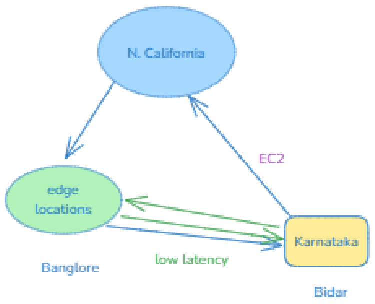
- These are big data centers present outside the region
- There are 13 REC's
- Whenever AWS is Unable to find the service in edge locations the request will be sent to REC's.

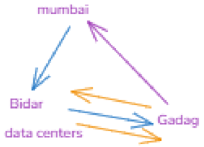
latency -> the time taken to transfer the data

low latency -> less time to transfer data

high latency -> more time to transfer data




Low latency --> high data speed
high latency --> low data speed

## AWS services across the various domain

1. Cloud financial management
   - eg: Billing and cost management
2. Compute
   - eg: EC2(Elastic compute cloud), Lambda
3. Storage:
   - eg: S3(Simple storage service), EBS(Elastic Block Storage)
4. Database:
   - eg: Aurora RDS(Relational Database service), DynamoDB
5. Networking service:
   - eg: VPC(Virtual Private Cloud), ELB(Elastic Load Balancer), Route53
6. App Integration:
   - eg: SNS(Simple Notification Service)
7. Security, Identity and Compliance
   - eg: IAM(Identity access management)
8. Management:
   - eg: CloudWatch

# AWS Global Infrastructure

The AWS Global Cloud Infrastructure is a secure, extensive, and reliable cloud infrastructure, offering over 200+ fully featured services from data centers globally. Whether you need to deploy your application workloads across the globe in a single step or you want to build and deploy specific applications closer to your end users with single-digit millisecond latency

**Components of AWS Global Infrastructure :**

1. Regions
2. Availability zone
3. Edge locations
4. Regional edge caches

***https://aws.amazon.com/about-aws/global-infrastructure***

1. ***Region***

AWS has the concept of a Region, which is a physical location around the world where we cluster data centers

***To Select region there are some parameters***

1. Data governance & Legal requirements.
2. Low latency
3. New features may or may not be available in all the regions.
4. Pricing may vary from one region to other region.

**2. *Availability Zone(AZ)***

An Availability Zone (AZ) is one or more discrete data centers with redundant power, networking, and connectivity in an AWS Region.

In a region (each and every region there are minimum 3 AZ's and maximum 6 AZ's

There are ~~108~~ 114 Az's present across the globe.

## Features of AWS:

1. Highly flexible
2. Pay as you go model
3. Highly secured
4. Scalable and elastic
5. Cost effective

## EC2(Elastic compute cloud)

Resizable

Storage,
CPU,
RAM

Over the internet

- AWS EC2 was launched in the year 2006
- EC2 comes under the category of Compute and follows the concept of IAAS
- EC2 is a cloud service offered by AWS, which provides on-demand, scalable, compute capacity over the internet.

## Components of EC2

1. Instances -> Virtual server/ virtual machine
2. EBS(Elastic block storage) -> Virtual hard drives
3. AMI(Amazon machine images)
4. ELB(Elastic Load Balancer)
5. ASG(Auto scaling group)

### 1. Instances:

To rent virtual machines/servers

Types of Instance

1. General purpose: Provides a balance of compute, memory(RAM), and network resources.
2. Compute-optimized: Suitable for application which requires large amount of CPU's
   - large amount of Processors
3. Memory-optimized : Huge amount of Memory (RAM)
   - to deliver fast performance that process large amount of data
4. Storage-optimized: Huge amount of storage
5. High-performance capacity(HPC): We'll get huge amount of CPU, RAM and storage
6. Accelerated Computing: to develop any applications involving AI and ML.

Pricing of EC2

per month -> 720 hrs is free

10 mins -> 1hr

1hr -> 1hr

2hr 30min -> 3hr

2 instance * 3 days => 6days (144hrs)

0.038 USD per hour

## Steps to Launch an instance in Linux/Ubuntu

1. Search for EC2 in AWS management console
2. Click on EC2
3. Click on launch instance
4. Give a name to your instance
5. Select the image (Linux/Ubuntu)
6. Select AMI(Which is under free tier)
7. Select instance type(t2.micro)
8. Create a new key pair or select existing key pair
9. Specify volume in configure storage and select gp2/gp3(min 8Gb and max 16384GB)
10. Click on launch instance

### Steps to connect instance (Linux/ubuntu)

1. Select the instance and click on connect
2. Select ec2 instance connect
3. User-name (ec2-user or ubuntu)

or

1. Select the instance and click on connect
2. Select ssh client and copy the example link
3. Do to file explorer and in the path of key pair change it to "cmd" and click on enter
4. Paste the copied ssh client link

NOTE: The minimum and maximum storage is 8GB and 16TB respectively

## Basic Linux commands

```
1. mkdir folder_name   --> to create a folder
2. cd folder_name  --> to get inside the folder or to change directory
3. cd .. --> to get to previous directory
4. touch file_name --> to create the file
5. nano file_name --> edit file or create file
6. vi file_name --> to edit or to create file
7. rm file_name  --> to delete file
8. rmdir folder_name --> to remove directory
9. ls  --> to list
10. cat file_name  --> to view the data of file
11. rm -r dir_name   --> to delete directory along with files in it
```

```
1. Create two folder
2. Create 2 files in each folder
3. in file1 of folder1 write the content as Hello all Welcome to AWS
```

## Steps to Launch an instance in Windows

1. Search for EC2 in AWS management console
2. Click on EC2
3. Click on launch instance
4. Give a name to your instance
5. Select the image Windows
6. Select AMI(Which is under free tier)(Microsoft server 2022 base (storage --> min 30GB and max 2048GB))
7. Select instance type(t2.micro)
8. Create a new key pair or select existing key pair
9. Specify volume in configure storage and select gp2/gp3
10. Click on launch instance

### Steps to connect instance (Windows)

1. Select the launched instance and click on connect
2. Select RDP client
3. Connection type -- connect using RDP client
4. Click on Download remote desktop file
5. Click on get password
   - Upload private key file(Key pair file)
   - Click on decrypt password
6. Copy the password
7. Open the downloaded remote desktop file
8. Give the password and make the connection.

# Elastic Block Storage

EBS is block storage service offered by AWS that provides reliable, high-performance storage for our EC2 instances.
It's like a virtual hard drive that you can connect to EC2 instance to store data
- Volumes are availability zone specific

2 components under EBS:

1. Volume: Additional storage for instance
2. Snapshot: It provides backup for volume and instance

### Steps to Create volume:

1. Go to AWS management console and search for EC2
2. Click on volumes under Elastic Block Storage
3. Click on Create volume
4. Select the volume type(GP2)
5. Give the size for the volume
6. Select the availability zone (same AZ as your instance)
7. Click on create volume

### Steps to Connect volume to Instance

1. Select the volume which you want to attach to instance
2. Click on actions and select attach volume
3. Select the instance which is in same AZ
4. Select the device name
5. Click on attach volume

1. Launch instance using Windows OS
2. Initialize volume
3. Launch instance in Linux and create 2 directories
4. Create a snapshot for that instance and create volume by using snapshot
5. Attach that volume to another instance
6. Attach 2 volumes for Windows OS

## Mounting volume on linux/ubuntu system

lsblk to list volumes
sudo file -s /dev/xvdf to check the file system
sudo mkfs -t ext4 /dev/xvdf to create a file system
sudo mkdir dir_name to create a directory
sudo mount /dev/xvdf dir_name to mount the volume

# Snapashot

1. launch linux/ubuntu instance
2. Create volume in same AZ
3. Attach that volume to instance
4. Mount the attached volume
5. Create a snapshot of the same volume

## Steps to create Snapshot for volume

- Click on create snapshot
- Click on volume
- Select the resource type(volume)
- Select the volumeID
- Give description and click on create snapshot

## Steps to create Snapshot for instance

- Launch instance and install software
- Click on create snapshot
- Click on Instance
- Select the resource(instance)
- Select the instance ID
- Give description and click on create snapshot

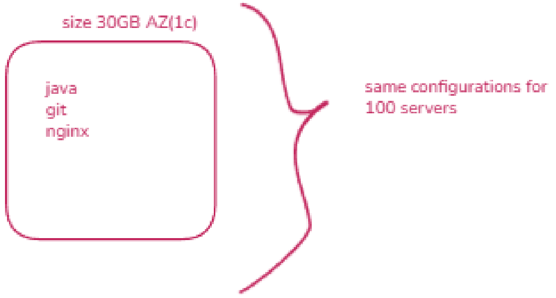
## Steps to create AMI using snapshot

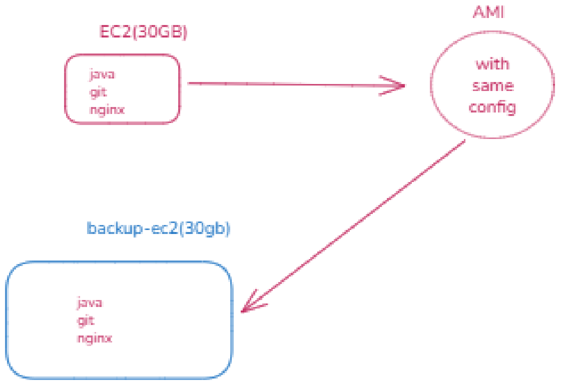
- Create snapshot
- Click on action in snapshot
- click on create image from snapshot
- Name the image and give description
- Click on create image

# AMI(Amazon Machine images)

- It is a blueprint for your instances, by using this you can launch multiple instances with same configuration
- It will provide a backup of software or application of your instance

ex: You want 100 servers in which by default java, git and nginx is installed with same configuration



## Steps to install software in Linux/Ubuntu

always update system before installing any softwares/tools

sudo yum update   Linux

sudo apt update   ubuntu

| Linux | ubuntu |
| --- | --- |
| sudo yum install java -y | sudo apt install openjdk-17-jdk |
| sudo yum install git -y | sudo apt install git -y |
| sudo yum install nginx -y | sudo apt install nginx -y |

1. updated linux system   sudo yum update
2. install java   sudo yum install java -y
3. Install git   sudo yum install git -y
4. install nginx   sudo yum install nginx -y
5. Check the status of nginx, whether it's active:   systemctl status nginx
6. To start nginx server:   sudo systemctl start nginx

## Nginx: It is a web server, a place to deploy static website

by default nginx port no. is 80

so make sure port no. 80 is open(enabled) in security group

default nginx path in amazon linux:  /usr/share/nginx/html

default nginx path in ubuntu:  /var/www/html

If nginx page is not loading the reason might be:

1. Port 80 might not be enabled under security group
2. when you are pasting public ip of instance, it might be in https
3. Nginx status might be stopped(dead/inactive)

## Steps to do static website hosting in Linux/Ubuntu

1. Update Linux/Ubuntu machine

   > sudo apt update

2. Install nginx and zip

   > sudo apt install nginx -y

   > sudo apt install zip -y

3. Download sample html project from Internet(Tooplate) and zip it into app.zip

   > wget -O app.zip  <project-download-url>

4. Unzip the zipped app.zip

   > unzip app.zip

5. Deploy the unzipped project in nginx server

   > sudo cp -r project-folder/*  /usr/share/nginx/html

created image from Instance

launch instance from AMI

## AMI(Amazon Machine images)

### Steps to create AMI for an instance using instance

- Launch an instance and install softwares in it
- Click on actions and select image and templates
- Select create image
- Name the image and give description
- Click on create image

## Drawbacks of EBS

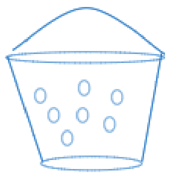
- It is availability zone specific
- Volumes can only be attached to the instances
- One volume cannot be connect or shared among multiple instances
- EBS have performance limits based on their type
- The maximum storage using EBS is 16TB
- to store small amount of data

# S3(Simple storage service)

- It was launched in the year 2006
- It comes under category of storage
- Amazon S3 is an storage service that offers industry-leading scalability, Data availability, security and performance.
- We can store and retrieve the data using the S3 through internet globally
- Provides unlimited storage
- S3 service is global specific
- To store large amount of data
- Components of S3

1. Bucket: Is a container for the object.
2. Object: It is a file or any data that describes a file.

use case of S3



s3 bucket

1. Storage
2. Backup
3. Huge files
4. Data activities
5. Static web hosting

object size: Maximum object size should be 5TB
10,000 free S3 bucket

## Steps to create bucket

1. Open AWS management console and search for S3
2. Click on "create bucket"
3. Name the bucket
4. ACL's ownership(disabled)
5. Block all public access(check)
6. Create bucket

## Steps to upload objects in S3 bucket

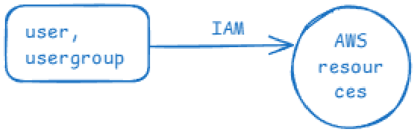
1. Open the bucket you created
2. Click on upload file
3. Select the files and folder or drag and drop files/folder tp upload
4. Click on upload

## Steps to host static website using S3

1. Create a bucket
2. Open the browser and go to "tooplate" website
3. Copy/download any html templated(folder)
4. Unzip the folder(extract)
5. Click on bucket
6. Upload objects to the bucket by clicking on add files/folders
7. Click on upload
8. Click on close
9. Go to permissions --> uncheck block all public access
10. Generate bucket policy
    - Click on edit in bucket policy
    - Copy ARN
    - click on policy generator
    - select type of a policy(S3 bucket policy)
    - principle(* -->all)
    - Select the action(getObject)
    - paste the arn (arn:aws:s3:::bucket_name)
    - click on add statement
    - click on generate policy
    - copy the generated policy
11. Paste the copied policy in bucket policy
12. click on save changes
13. go to properties --> click on edit in static website hosting
    - Click on enable
    - Give the file name(index.html)
    - Save changes
14. Go to browser and paste endpoint(static website hosting link)

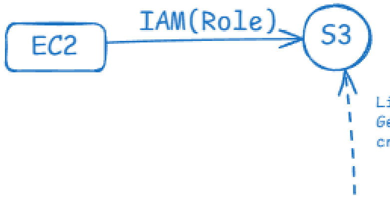
# IAM(Identity and access management)

- It comes under the category of security, Identity and compliance
- IAM is a web service that helps you securely control access to AWS resources
- We can manage permissions that control which AWS resources can users access.




## 4 components of IAM

1. Users
2. User group(Group of users with common permissions)
3. Policies(Set of permissions)
4. Roles(Set of policies)

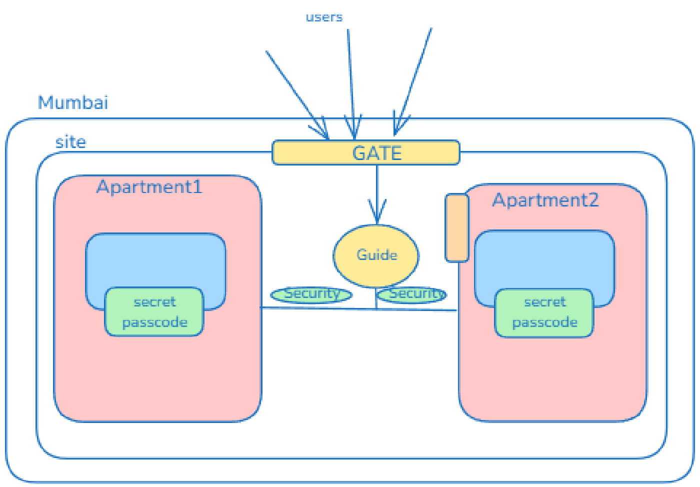


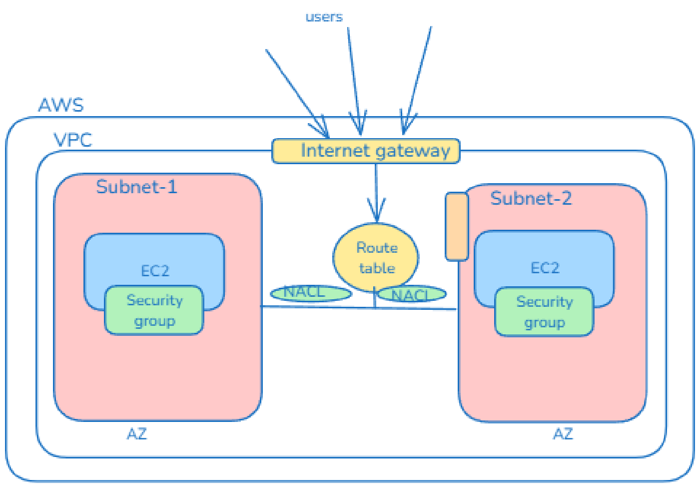

ListBucket, ListAllMyBucket
GetObject
createBucket, DeleteObject, DeleteBucket, putObject

# VPC(Virtual private cloud)

- VPC is to create a own private cloud within public cloud

- It uses IAAS concept




Internet gateway: It is to communicate with internet and AWS resources

Subnet: It is a small portion of VPC, it is used to isolate the resources

2 types of subnet:

1. Public subnet: to communicate with internet
2. Private subnet: Connected to NAT
   by default only outbound is possible

Route table: It will direct the user to the subnets

NACL(Network access control list): It will handle inbound and outbound in subnet level.

Security groups: It will handle inbound and outbound in instance level

NAT gateways: It is private gateway for your subnets which will only allow outbound by default.

## Address:

### Physical address

- Hardware devices.
- It is static
- Called as mac address
- Not discoverable over the internet

> getmac

### Logical address

- Generated by the software
- It is dynamic
- called as IP address
- Discoverable over the internet

> ipconfig

## IP address

IPv4 address
(Internet protocol version 4)

IPv6 address
(Internet protocol version 6)

## 1. IPv4 address:

___.___.___.___

Four set of numbers
each seperated by dots(.)

0-255

- 256 possibilities for each set
- 256^4 = 4,29,49,67,296
- 4.3 billion total IPv4 address are possible
- 0.0.0.0 to 255.255.255.255
- Large number of IP address are not possible (disadvantage)

Class A:

0.0.0.0 to 126.255.255.255
- used in larger network

0.0.0.0
0.0.0.1
0.0.0.2
......
....
....
0.0.0.255
0.0.1.0
....
126.255.255.255

Class B:

128.0.0.0 to 191.255.255.255
- used in Medium scaled organization

128.0.0.0
128.0.0.1
128.0.0.2
......
....
....
128.0.0.255
128.0.1.0
....
191.255.255.255

Class C:

192.0.0.0 to 223.255.255.255
- used in small scaled organization

Class D:

224.0.0.0 to 239.255.255.255

used for multi-screening

Class E:

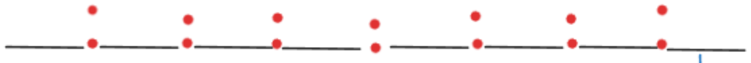
240.0.0.0 to 255.255.255.255

used for testing or research

loopback

127.0.0.0 to 127.255.255.255

## 2. IPv6 address:



There are 8 set of alphanumeric numbers
each seperated by ":"

16^4 = 65536

65536^8 = 3.4 *10^38

16 possibilities

0-9 numbers
a-f alphabets

## IP address:

### Public IP address

- It is unique over the internet
- Accessible using browser
- To communicate over the internet

### Private IP address

- It is unique within the network
- Not accessible
- for internal communication within the network

# VPC creation

- VPC it is an isolated, customizable network within AWS where you create and manage the services

total IPv4 = $\frac{429,49,67,296}{2^{24}}$ = 256

CIDR(Classless Inter-Domain routing)

Defines IP address range

0-32

$\frac{256}{4}$ = 64

| subnet-1 | subnet-2 | subnet-3 | subnet-4 |
|---|---|---|---|
| 10.0.0.0 | 10.0.0.64 | 10.0.0.128 | 10.0.0.192 |
| 10.0.0.1 | 10.0.0.65 | 10.0.0.129 | 10.0.0.193 |
| 10.0.0.2 | 10.0.0.66 | 10.0.0.130 | 10.0.0.194 |
| .. | .. | .. | .. |
| .. | .. | .. | .. |
| .. | .. | .. | .. |
| .. | .. | .. | .. |
| .. | .. | .. | .. |
| 10.0.0.63 | 10.0.0.127 | 10.0.0.191 | 10.0.0.255 |
| 10.0.0.0/26 = 64 | 10.0.0.64/26 = 64 | 10.0.0.128/26 = 64 | 10.0.0.192/26 = 64 |

VPC creation → VPC and more

VPC creation → VPC only

VPC and more

- VPC, subnets, Internet gateway, route tables, NAT

## Steps to create VPC (VPC and more)

1. Go to AWS management console and search for VPC and select it
2. Select your VPC and click on create VPC
3. Resources to create --> VPC and more
4. Name your VPC
5. Select CIDR(10.0.0.0/24 ->256IPv4), select the IP address
6. Select number of Availability Zone(2)
7. Number of Private and public subnet(2,2)
8. Select NAT gateway(in 1 AZ)
9. Endpoints(none)

## Steps to create EC2 Instance in VPC

1. Go to EC2 dashboard
2. While creating instance edit the network settinngs
3. Select the VPC and Availability Zone
4. auto-assign public IP -> enabled
5. Select the security group or create it
6. Create instance

## Steps to modify inbound rules in security group

1. Go to EC2 dashboard
2. Click on security groups
3. Select the security group
4. Select the inbound rule
5. Edit inbound rules(add or remove rules)
6. Save it

## Resources to delete before deleting VPC

1. Terminate the instance
2. Delete the NAT gateway
3. Release elastic IP's
4. Delete VPC

VPC only

ne

# Steps to create VPC using VPC only

1. Click on create VPC
2. Select VPC only
3. Name your VPC
4. Select CIDR block
5. create VPC
6. Create four subnets(2private and 2public)

   steps to create subnet

   - Go to subnets
   - Click on create subnet
   - Select the VPC
   - Select CIDR block for subnet
   - Add a subnet by clicking on add subnet

7. Create 3 route tables(2 for private subnet and 1 for public subnet)

   steps to create route tables

   - Go to route tables
   - Click on create route table
   - Name the route table
   - Select the VPC
   - Create route table

8. Attach the route table with subnets

   Steps to attach the route table with subnets

   - Select the route table
   - Click on actions
   - Edit subnet associations
   - Select the subnet
   - Save

9. Create internet gateway for public subnet

   Steps to create internet gateway

   - Go to internet gateway
   - Create internet gateway
   - Name it
   - Create it
   - Select the gateway and attach it to VPC by clicking on actions

10. Attach internet gateway to public route table

    Steps to attach route table to internet gateway

    - Select the route table
    - Click on actions
    - Select edit routes
    - Add route
    - Select 0.0.0.0/0 destination
    - Select internet gateway in resource type
    - Save

11. Create NAT gateway

    Steps to create NAT gateway

    - Go to NAT gateway
    - Click on create NAT gateway
    - Select public subnet
    - Create

12. Attach NAT gateway to route table

    Steps to attach route table to NAT gateway

    - select the route table
    - click on actions
    - Select edit routes
    - Add route
    - Select 0.0.0.0/0 in destination
    - select the NAT gateway in resource type
    - Save

# Lambda



- It is serverless compute service that provides the platform to run program or code

| EC2 | Lambda |
|---|---|
| - Need to create and manage the server.<br>- For scaling you have to setup auto scaling.<br>- Pay for running server even though you are not using it | - No need to create and manage the server.<br>- Automatically scale based on request.<br>- You will pay the computing time you use. |

## Pricing of lambda

1. Depends on the request.
   1 million free requests can be done per month under free-tier.

2. Depends on computational time.

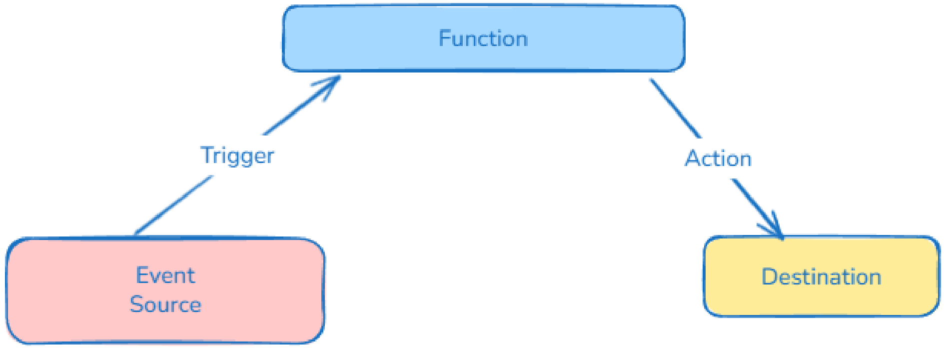
   Costing : GB*second

   100GB * 5sec = 500 GB-sec

   4 lakh GB-sec (free-tier)

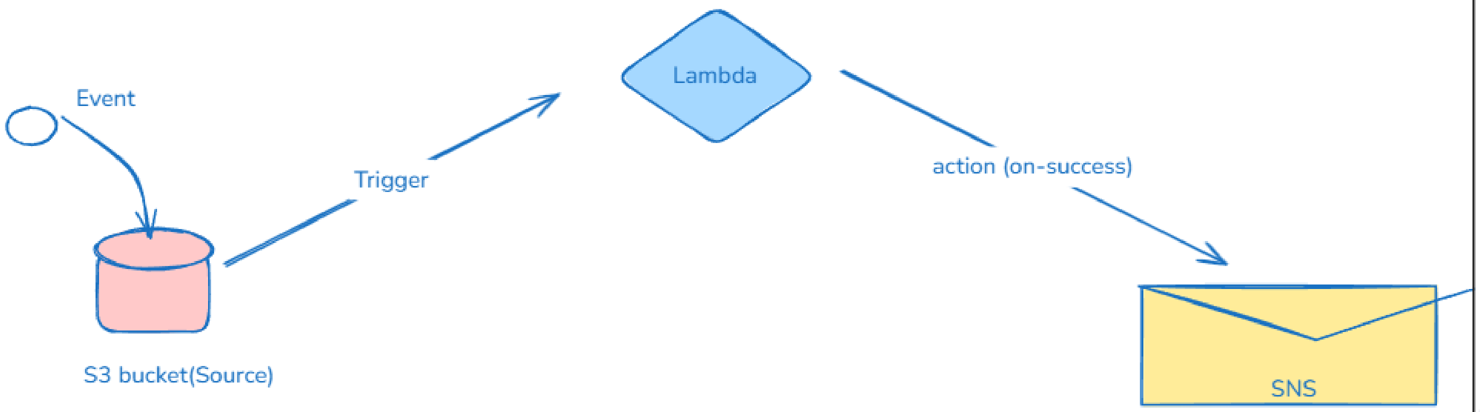
## Structure of Lambda

1. Functions
2. Event Source(Triger)
3. Destination(Action)




Event source:

Any type of request that triggers a lambda to run.



Types of Event:

1. Push-based event
2. Pull-event based : It will check for new data in a source and run the functions when new data is found

Push-based events:

lambda is triggered by an action

Types of push-based event:

- Synchronous: Events wait for lambda to execute the task before triggering it again
- Asynchronous: Events will trigger the lambda immediately.

## Script to start stopped instance:

```
import boto3
region ='region'
instances =['instance id' ]
ec2=boto3.client('ec2',region_name=region)

def lambda_handler(event,context):
    ec2.start_instances(InstanceIds=instances)
    print('Started your instances:'+str(instances))
```

## Script to stop running instance:

```
import boto3
region ='region'
instances =['instance id' ]
ec2=boto3.client('ec2',region_name=region)

def lambda_handler(event,context):
    ec2.stop_instances(InstanceIds=instances)
    print('Stopped your instances:'+str(instances))
```

## Steps to create lambda functions:

1. Search for lambda in AWS management console
2. Select functions and click on create function
3. Name your functions and select runtime(python 3.13)
4. Create function
5. Paste the script and replace the region and instance id(edit script according to request)
6. Click on test and create event
7. Configure the permissions
8. Click on test(your program/ function will run)
9. Click on deploy

# SNS (Simple notification service)

Subscriber —subscribe→ apnaCollege

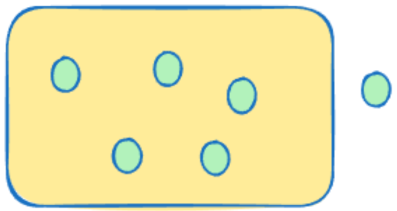
youTube(channel)

- Comes under the category of app integration
- It is managed notification service in AWS
- It supports e-mail, http, SMS
- Provides message delivery from producers(publishers) to consumers(subscribers)

Components of SNS:

Topic:

- Defines the end-point(whom) for the notification
- subscribing endpoint to it



Subscriber

Topic

## Steps to create a SNS topic

1. Search for SNS in AWS management console and select it
2. Click on create topic
3. Name the topic
4. Create a subscriptions
   - Click on create subscription
   - Select the protocol as e-mail
   - Give the e-mail I'd and save it

## Steps to add S3 as a trigger for lambda

1. Create a function(author from scratch)
2. Click on add trigger
3. Select S3, select the bucket and event as put
4. Save

## Steps to attach SNS to lambda function

1. Go to lambda function
2. Click on add destination
3. Select SNS topic
4. Select the topic name
5. Save

Canary function -> blueprint

website (www.qspiders.com) —on-failure (2 minutes)→ searching for a key word (AWS)

website (www.qspiders.com) —on-success (2 minutes)→ searching for a key word (Training)

, -> multiple values (15,30,45)
- -> range (10-15)
* -> every minute
/ -> every per (2/10)

To move canvas, hold mouse wheel or spacebar while dragging, or use the hand tool

# CloudWatch

- CloudWatch comes under the category of management
- It is a monitoring service offered by AWS

**Components of CloudWatch**

- 1. Dashboard
  2. Metrics
  3. Alarms

1. Dashboard:

Using Dashboard we can analyze the resources based on metrics.

types of Dashboard:

1. Automatic dashboard
2. Custom dashboard

2. Metrics: To measure the substances

Vegetables -> kg, grams
Milk -> litres, ml
height -> cm, metre, inches, foot

In AWS to measure each and every resources we have multiple metrics
eg: In EC2 instance - cpu utilization

Widgets: To analyze metrics in different formats.
eg: Number, Line-graph, pie-chart, bar-graph

3. Alarms: time ------> 5:00Am --------> Ring
metrics condition action

In EC2 instance:

CPU utilization --> less than 10% ---> terminate the instance

stop instance
restart instance
terminate instance

Steps to create an alarm that stops the instance if CPU utilization is less than 10%

1. Go to AWS management console and search for CloudWatch
2. In the cloudwatch dashboard select in alarms
3. Click on create alarm
4. Click on select metrics and click on EC2
5. Click on per-instance metrics
6. Copy the instance id and paste it in the search bar
7. Select the metric(CPU utilization)
8. Click on select metric
9. Select the statistics(Average)
10. Select the period(custom --2 minutes)
11. Threshold type --> static
12. Whenever the CPU Utilization is -- lower than ---10
13. Click on next
14. Alarm state trigger(in alarm)
15. Select the SNS topic to create function
16. Go to EC2 action -- add EC2 action
17. Alarm state trigger --> in alarm
18. Select the action --> stop instance
19. Click on next
20. Give the alarm name and click on next.

create 4 dashboards for same instance but with different widgets

-> create 2 instances
-> in VM1 terminate the instance if CPU utilization is less than 10%
-> Stop the VM2 if CPU utilization is less than 20%

## ASG(Auto Scaling Groups)

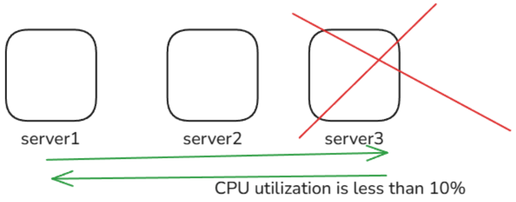
Scaling: Increasing and decreasing the number of resource

ASG: It is a scaling service offered by AWS, to automatically scale-up or scale-down the resources.

2 types of Scaling:

1. Horizontal scaling: Scaling based on number of resources.

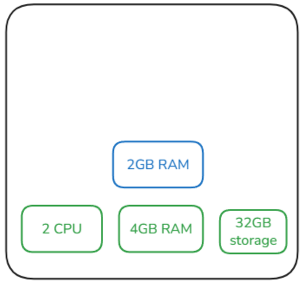
3-10




Condition : CPU utilization
(If the cpu utilization is greater than 80, new instance should be created automatically)

2. Vertical Scaling: Scaling based on configuration of resources

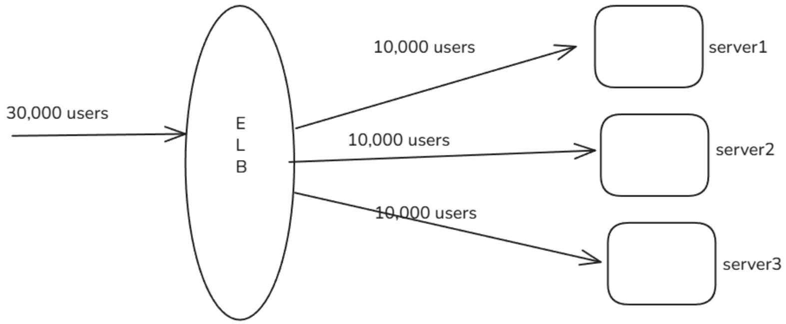
eg: Storage, CPU, Memory




server1

## ELB(Elastic Load Balancer):

This will divide the loads equally among multiple servers




1. launch ubuntu instance

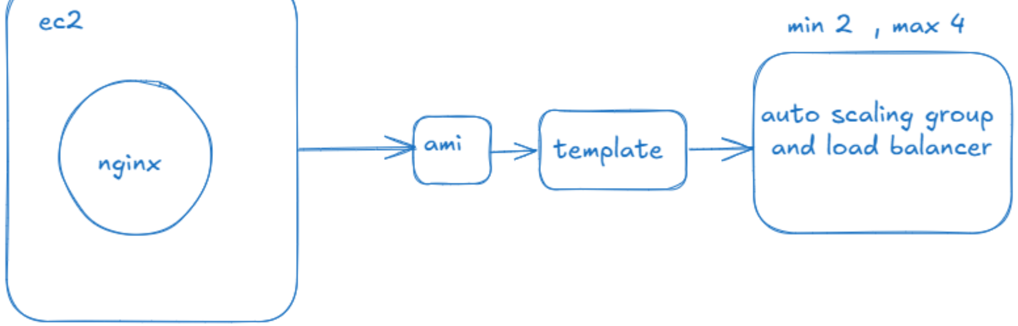
-> install nginx
-> do static web site hosting

NOTE: port 80 should be allowed in security group

2. Create Image of the instance
3. Launch a template using the image

4. ASG and ELB

-> create ASG
-> Integrate load balancer
-> create a target group




steps for auto scaling

1) create an ec2 instance and launch a website
2) create an AMI using that instance
3) create a launch template
   - launch template name
   - check Auto Scaling guidance
   - select the ami
   - instance type
   - key pair
   - don't select the subnet
   - select the security group
   - create launch template

4) create auto scaling group
   - click on create auto scaling group
   - give name and select the launch template ---> next
   - select vpc and subnets --> next
   - attach with new load balancer
   - give name and select application load balancer
   - select internet facing
   - create the target group --> next
   - select desired capacity
   - select min and ma capacity
   - select target traking
   - select cpu utilization metric and setup --> next
   - select sns if you want
   - next
   - create asg

## Database:

It is a place where we store the data

To create database we need:

1. DBMS
2. Query language
3. Host

DBMS(Database management System):

It is used to create, retrieve, manage and monitor the data in the database using the Query language.

Types of DBMS:

1. RDBMS: The data will be stored in table format i.e., rows and columns

   eg: MySQL, PostgreSQL, Aurora SQL, AWS RDS, OracleSQL

2. Non-Relational DBMS: Used to store unstructured or semi-structured data
   - We, store the data in key and value pairs

   eg: MongoDB, DynamoDB, Redis

Query Language(QL):

It is the language to communicate with database

eg: SQL, NoSQL

Host: Machine where you store the data

eg: Server, Laptop

# RDS(Relational Database service)

**RDS is a managed database service offered by AWS that supports various database like MySQL, PostgreSQL, OracleSQL**

## Benefits/Advantages:

1. Easy-setup
2. Backup
3. Scaling
4. Security
5. High-availability

### Components of RDS:

1. Database engine
2. Database Instance
3. Database cluster
4. Security group
5. Subnet group

MySql Installer
MySQL workbench 8.0

MySQL port number = 3306

numbers => integer
string => varchar(50)

### Steps to create and manage RDS:

1. Go to AWS management console and search for RDS
2. In RDS dashboard click on databases.
3. Click on create database
4. Select standard create.
5. Select database engine type as MySQL
6. select engine version (8.0.41)
7. Choose the templates (free tier)
8. Unser setting:
   - give db instance name
   - give master username(admin)
   - give user password(Admin1234)
9. Configure storage(20Gb)
10. Connectivity (don't connect with EC2 compute resource)
11. Public Access (yes)
12. Click on "create database"
13. Copy endpoint of database
14. Open MySQL workbench and make connection by pasting endpoint
    - click on "+" symbol
    - give connection name (demo)
    - connection type(standard TCP/IP)
    - give hostname(copied endpoint)
    - give username(the name while create db instance)
    - give password by clicking on store vault (the one while creating the db instance)
    - click on test connection
    - click on ok

# DynamoDB

- It is under the category of database
- Dynamo is noSQL database which is used to store the data in unstructured format.
- It is serverless database

# AWS CLI

- Managing the AWS resources by executing commands is called as AWS CLI (Command line interface)

3 Ways to access AWS service

1. AWS management console
2. AWS CLI
3. SDK(Software development kits)

to achieve AWS CLI:

1. Create IAM user - attach policy to use resources
2. Inside user, click on security credentials
   -> click on edit in "access key"
   -> give use case as "AWS CLI" and create access key
   -> copy "Access key" and "Secret Key"

3. Open command prompt
   -> aws --version
   if you are not getting version of AWS then install AWS CLI
4. Install AWS CLI
   -> Open browser and search for "aws download for windows"
   -> install AWS CLI

5. Go to command prompt and use CLI
   -> aws --version
   -> aws configure
   -> Access Key:  paste-cpoied-access-key
   -> secret key: paste-copied-secret-key
   -> default region name(us-west-1c): ap-south-1
   -> output format(none):
   -> start using aws cli from here

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